

Robotic Arm

Section: P1

Group No. 5

Group Leader: Vivek Gunuganti (20230395)

Other members: Pulkit Maheshwari (20230413), Raj Vardhan Jain (20230433), Shivansh Srivastava (20221124), Shobhit Methi (20230469), Ayush Aman (20230228), Divyansh Bohara (20230448), Himanshu Gupta (20230160), NageswaraSatyaShirdinadh Avupati (20230446), and Harshit Jaiswal (20231286).

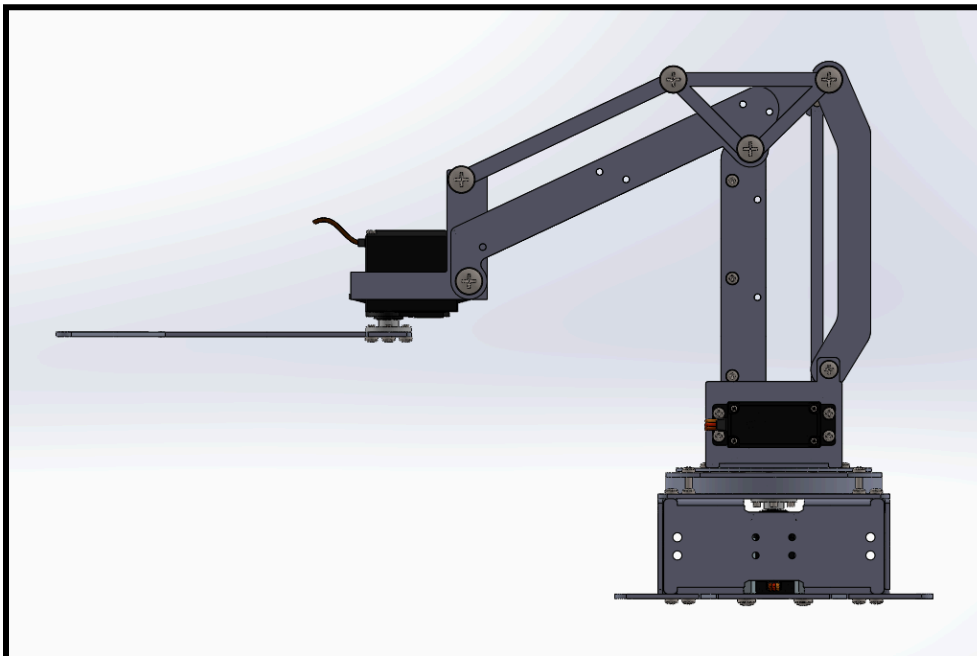
1. **Introduction:** This project focuses on designing, developing, and manufacturing a robotic manipulator for industrial automation. The robotic arm will be engineered for precision, flexibility, and efficiency, catering to applications such as assembly, material handling, and pick-and-place operations.

2. **CAD Design:**

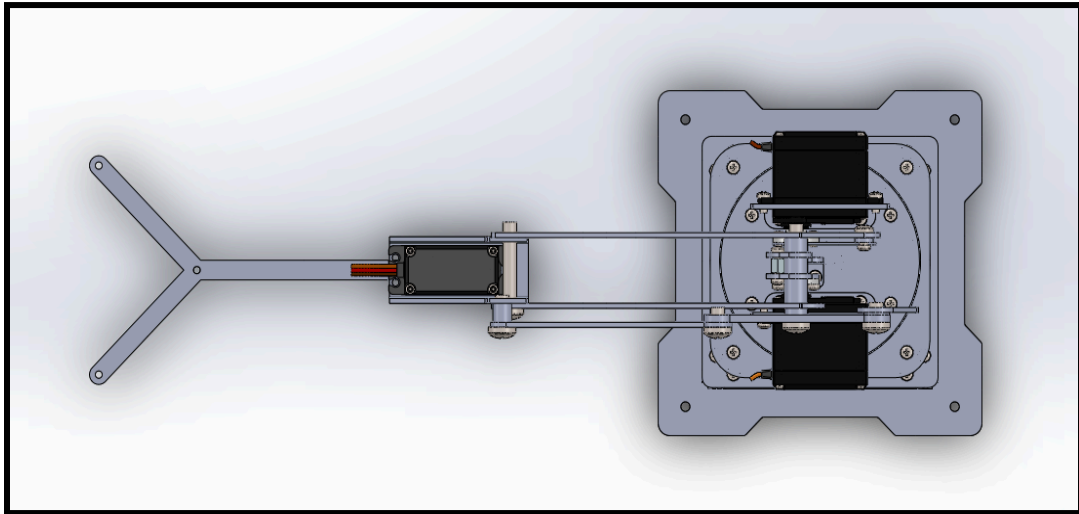
Assembly_Videos:

https://drive.google.com/drive/folders/1t2eTLGanw_q-xP1X9Zex6FVyx89WHU4U?usp=sharing

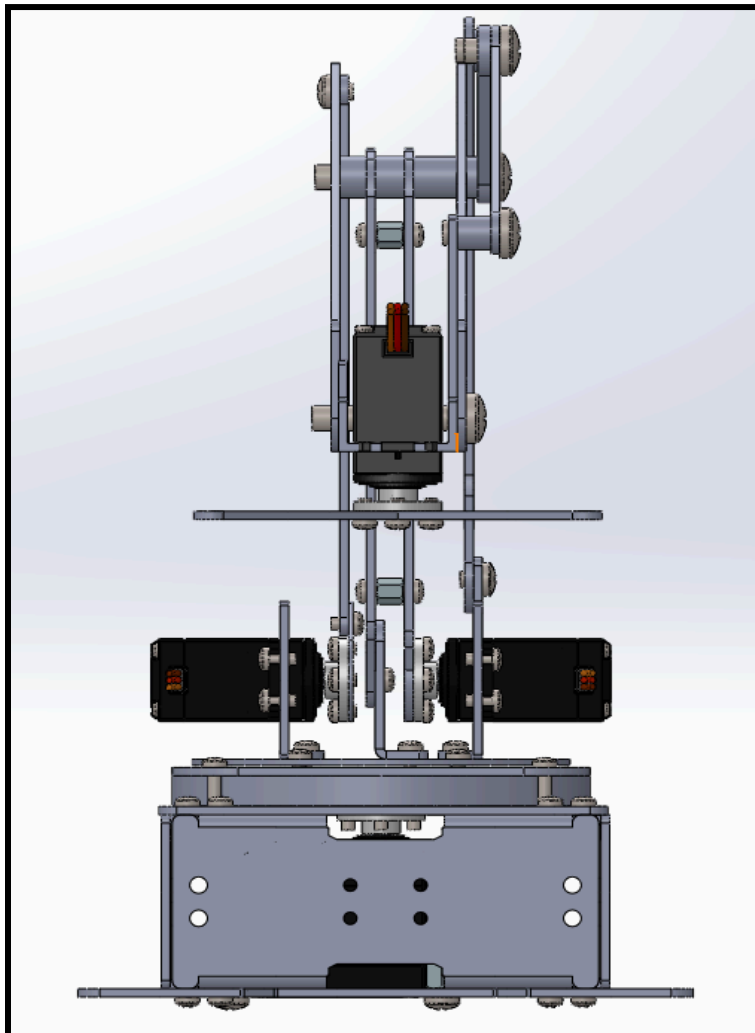
Side View:



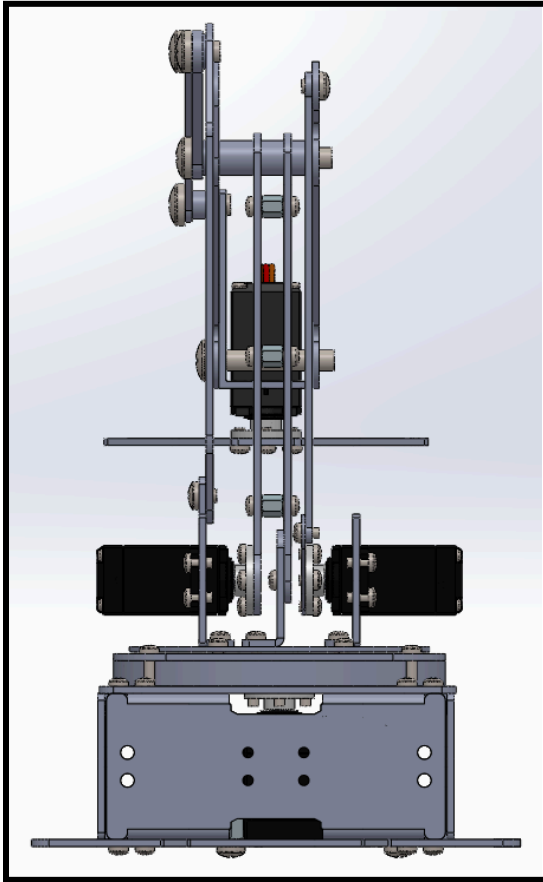
Top View:



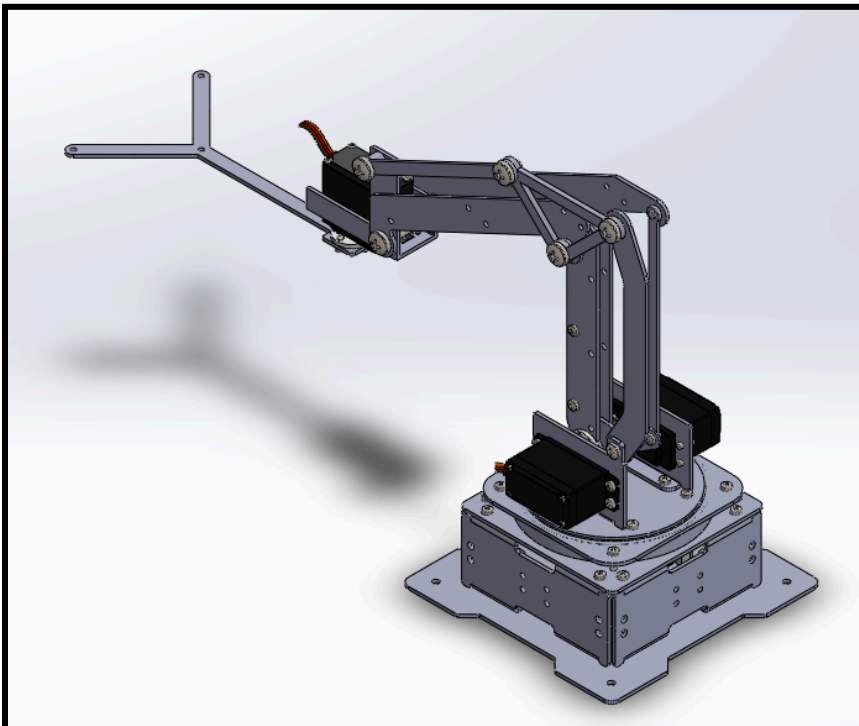
Front View:



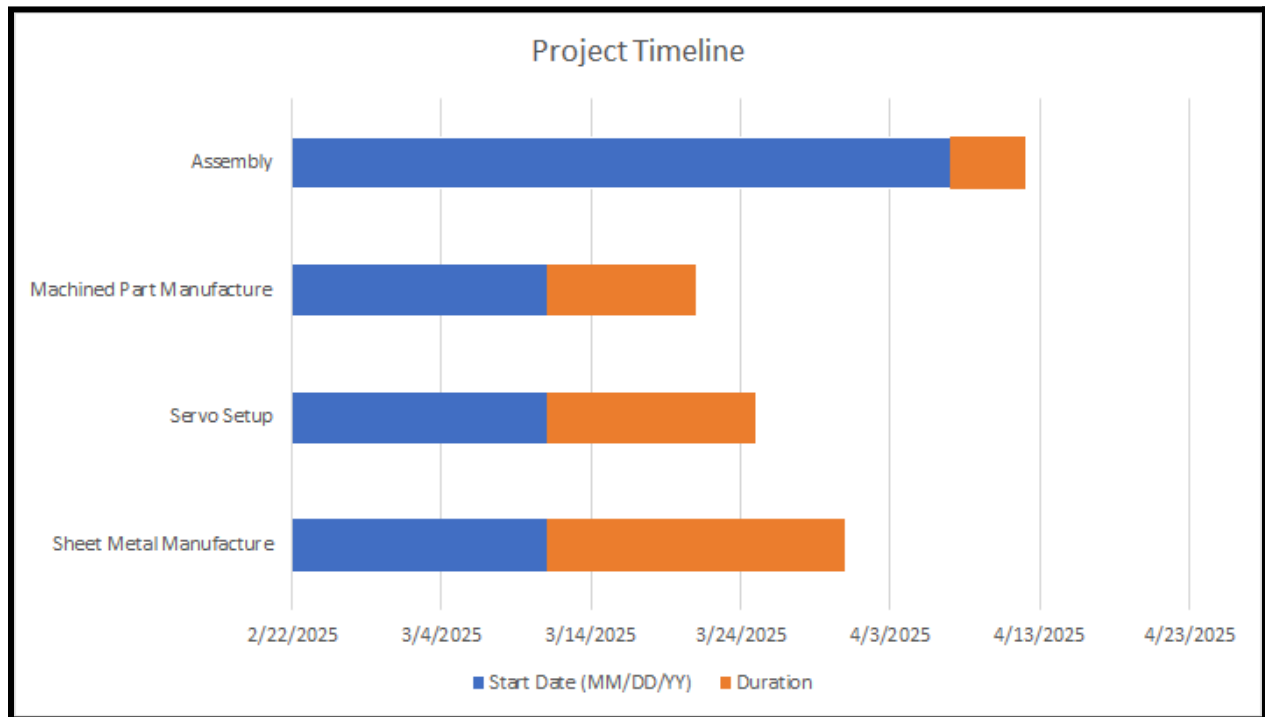
Back View:



Isometric View:



3. Work plan:



4. Brief description of functionality and application of the part

The robotic manipulator is designed for precision handling and high-accuracy assembly tasks:

Key Functionalities:

- 1. High-Precision Motion Control:**
 - Uses servo motors and encoders for sub-millimeter accuracy.
 - Smooth trajectory planning for delicate operations.
- 2. Fine Handling & Dexterity:**
 - Equipped with multi-fingered or adaptive grippers for secure yet gentle grasping.
 - Force and torque sensors enable real-time pressure adjustments to handle fragile components.

The robotic manipulator ensures efficiency, consistency, and high-quality output in fine assembly and delicate handling tasks.

Bill of Materials

Project title: Miniature Fine Assembly Robotic Arm

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Details of materials:

Item no.	Part Number	Material	Description	Qty.
1	Base_Part_Bottom	Aluminium	Primary Base of the Model	1
2	Base_Part_Side	Aluminium	Placed along the Sides of the Model to raise the Arm Platform	4
3	Base_Part_Top	Aluminium	Placed on the Bottom Bearing Platform to Allow Smooth Rotation	1
4	skf_bearing_61815	Aluminium	Used to Allow Rotation of the Arm	1
5	Theta_Mount	Aluminium	To hold the Base Servo	1
6	Bearing_Bottom	Aluminium	To hold the Bearing	1
7	Bearing_Top	Aluminium	To hold the Bearing	1
8	Bearing_Plate	Aluminium	To hold the Upper Segment of the Base	1
9	Motor_Clamp	Aluminium	To hold the crane servo	1
10	Arm_Mount	Aluminium	To mount Arm_1_A_B	1
11	Actuator_Arm	Aluminium	To connect the lifting lever with one of the servos	1
12	Motor_Clamp_2	Aluminium	To hold the second crane servo	1
13	Lifting_Lever_1	Aluminium	To constrain the motion of lifting lever 2_2	1
14	Bell_Crank	Aluminium	To constrain the motion of the arm 1_A	1
15	Lifting_Lever_2	Aluminium	To constrain the Motion of Arm 2_1_A	1
16	Lifting_Lever_2_2	Aluminium	To preserve the Orientation of the End Effector Servo Mount	1
17	Arm_2_1_A	Aluminium	Component of the Arm assembly, which in	1

			combination with 2_1_A_B gives one degree of freedom.	
18	Arm_1_A	Aluminium	Component of the Arm assembly, which in combination with 1_A_B gives one degree of freedom.	1
19	Arm_1_A_B	Aluminium	Component of the Arm assembly, which in combination with 1_A gives one degree of freedom.	1
20	Arm_2_1_A_B	Aluminium	Component of the Arm assembly, which in combination with 2_1_A gives one degree of freedom.	1
21	End_Effector_Servo_Platform	Aluminium	Houses the end effector servo	1
22	End_Effector	Aluminium	End Effector for specific tasks	1
23	Spacer_1	Aluminium	Used to preserve distances between arm components	1
24	Spacer_2	Aluminium	Used to preserve distances between arm components	1
25	Spacer_3	Aluminium	Used to preserve distances between arm components	1
26	Spacer_4	Aluminium	Used to preserve distances between arm components	1
27	Spacer_5	Aluminium	Used to preserve distances between arm components	1
28	Spacer_6	Aluminium	Used to preserve distances between arm components	1
29	Pan Head Screw: M4 x 0.7 x 8	Stainless Steel	Fasteners	4
30	Pan Head Screw: M3 x 0.5 x 6	Stainless Steel	Fasteners	28
31	Pan Head Screw: M3 x 0.5 x 16	Stainless Steel	Fasteners	8
32	Pan Head Screw: M4 x 0.7 x 6	Stainless Steel	Fasteners	4
33	Pan Head Screw: M3 x 0.5 x 5	Stainless Steel	Fasteners	11
34	Pan Head Screw: M3 x 0.5 x 4	Stainless Steel	Fasteners	15
35	Pan Head Screw: M4 x 0.7 x 5	Stainless Steel	Fasteners	2
36	Pan Head Screw: M6 x 1.0 x 40	Stainless Steel	Fasteners	1
37	Pan Head Screw: M6 x 1.0 x 8	Stainless Steel	Fasteners	2
38	Pan Head Screw: M6 x 1.0 x 35	Stainless Steel	Fasteners	1

39	Pan Head Screw: M6 x 1.0 x 13	Stainless Steel	Fasteners	1
40	M3 Washers	Stainless Steel	Fasteners	30
41	M4 Washers	Stainless Steel	Fasteners	12
42	M6 Washers	Stainless Steel	Fasteners	5
43	Standoff (M3x6)	Stainless Steel	Fasteners	3
44	Standoff (M4x28)	Stainless Steel	Fasteners	4

Details of processes:

S. No.	Name of Lab	Instructor	Remarks by instructors	Signature of instructor
1	Metal Cutting shop	Mr. Vinod Kumar Mr. Rakesh Kumar		
2	Forming shop	Mr. Sukhwant Singh Mr. Rajendra Kumar		
5	Lathe shop	Mr. Rajesh Kumar Tanwar		
6	Shaper, milling and grinding shop	Mr. Raj Kumar Saini		

Involved manufacturing processes (at least two): Yes/No

- 1. Metal Cutting (For individual parts of the robotic arm)**
- 2. Forming (For the bends in specific components)**
- 3. Lathe (For spacers, if required)**
- 4. Milling (For drilling holes)**

Approval of Workshop Floor In-charge

Approval of Mentor /IC